SlyLED User Manual

3D Volumetric Lighting System — v8.3.0

Complete guide to designing, programming, and running LED + DMX lighting shows with the SlyLED orchestrator, performers, and DMX bridge.

electricrv.ca/slyled

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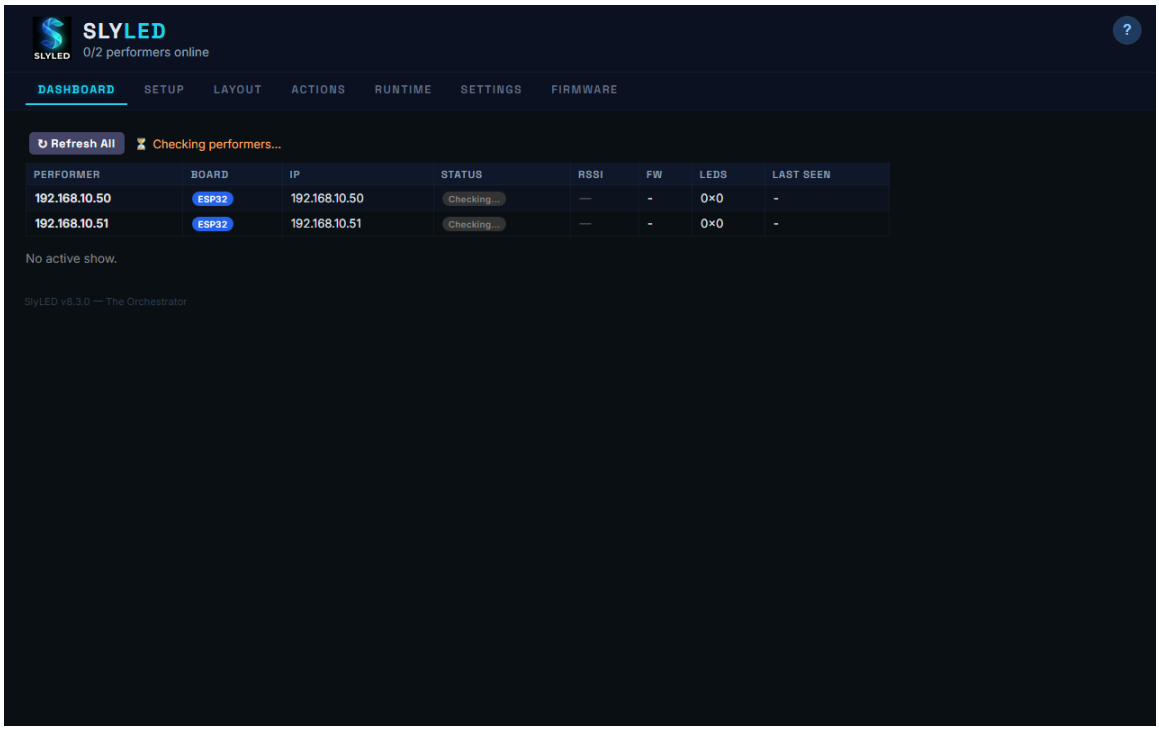
1. Getting Started with 3D Stage Design

Overview

SlyLED is a three-tier lighting system: the Orchestrator (Windows/Mac desktop app) designs and controls shows, Performers (ESP32/D1 Mini boards with LED strings) execute lighting effects, and DMX Bridges (Giga R1 boards) drive professional DMX fixtures over Art-Net.

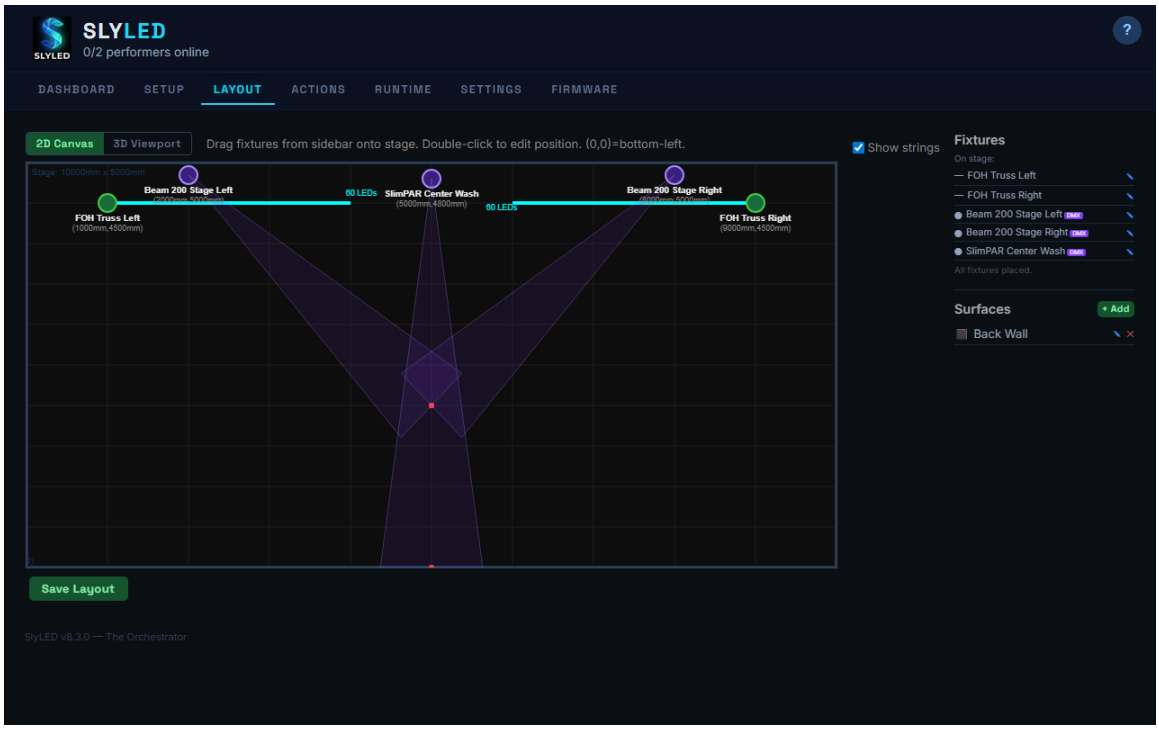


Dashboard — live performer status overview

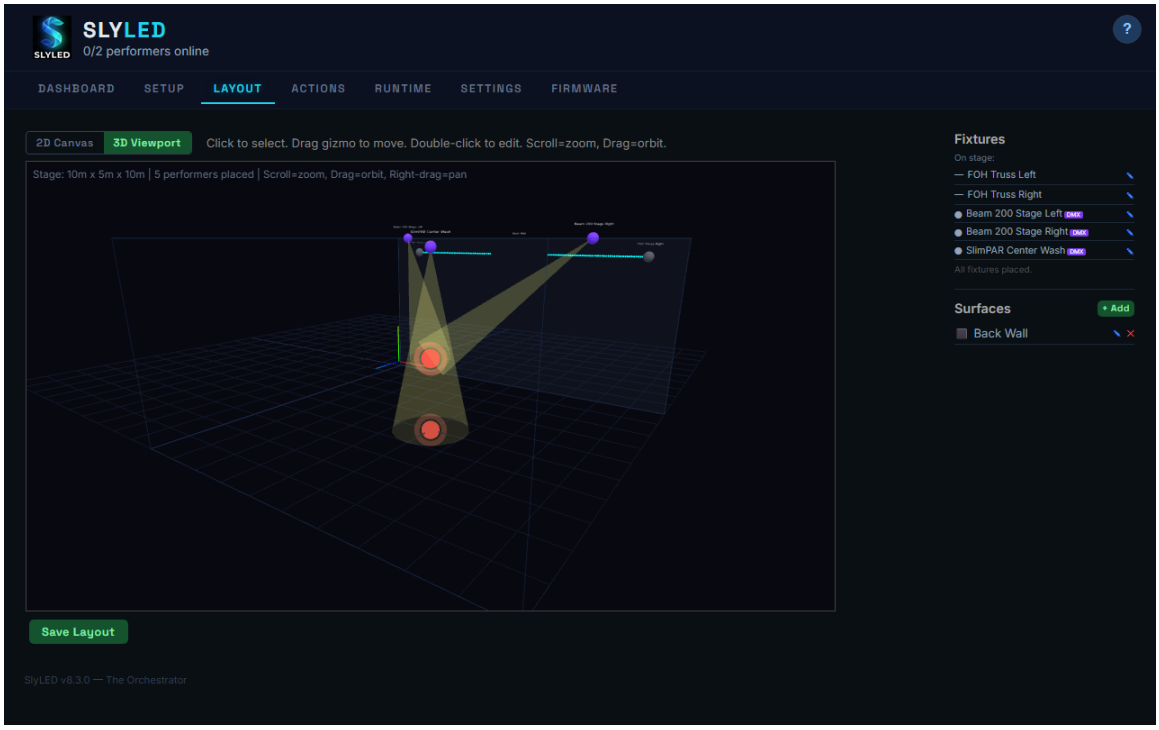
Switching Between 2D and 3D

The Layout tab offers two views:
- 2D Canvas: Flat top-down layout for simple setups
- 3D Viewport: Interactive Three.js scene for complex multi-level installations

Both views share the same position data. Switching is instant and non-destructive.



Layout tab — 2D Canvas with placed fixtures



Layout tab — 3D Viewport with stage wireframe, fixtures, and surfaces

Navigating the 3D Viewport

Action	Control

Orbit (rotate)	Left-click + drag
Zoom	Scroll wheel
Pan (shift view)	Right-click + drag
Select fixture	Left-click on node
Move fixture	Drag the colored arrows
Edit fixture	Double-click on node
Place from sidebar	Drag unplaced fixture into viewport

Coordinate System

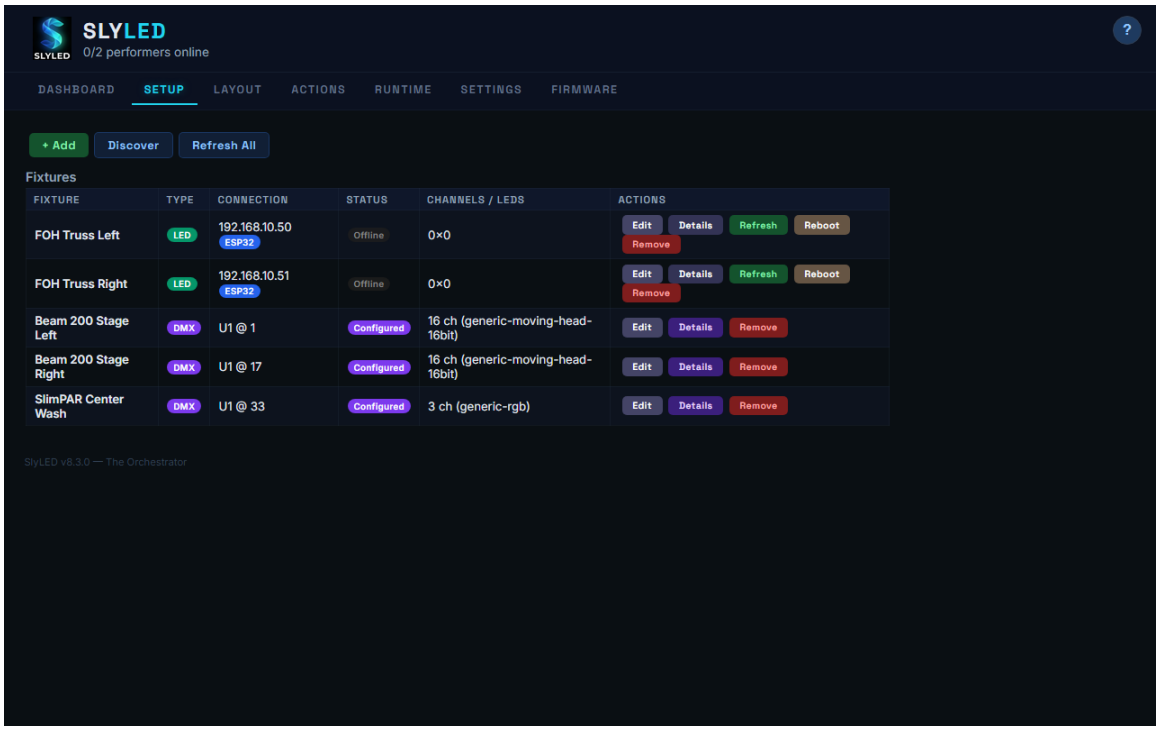
- X-axis (red): Width — left to right
- Y-axis (green): Height — ground to ceiling
- Z-axis (blue): Depth — front to back
- Origin: Bottom-left-front corner of the stage
- Units: Internally millimeters; displayed in Settings-chosen unit

2. Fixture Setup

What Are Fixtures?

A fixture is the primary entity on the 3D stage. It wraps physical hardware and adds stage-level attributes like position, rotation, and aim point.

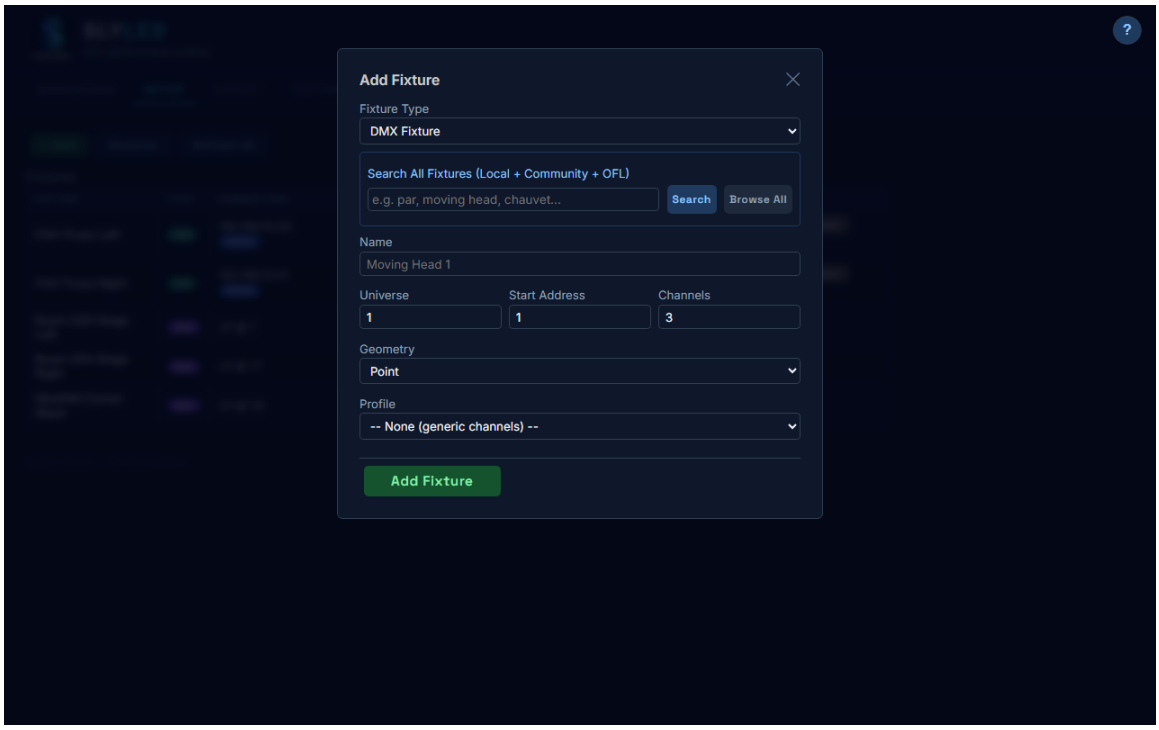
- LED Performers: Auto-created when hardware is registered via Setup tab
- DMX Fixtures: Created manually with universe, address, and profile assignment
- Fixtures can override child attributes (e.g., rotate a horizontal string to vertical)
- The baking engine uses the fixture's position and rotation, not the child's raw config



Setup tab — LED and DMX fixtures with type badges and status

Adding an LED Performer

Click Add Fixture, select "SlyLED Performer (LED)", and enter the device IP address. The system probes the device via UDP PING and HTTP, registers it as a child, and auto-creates a linked fixture.



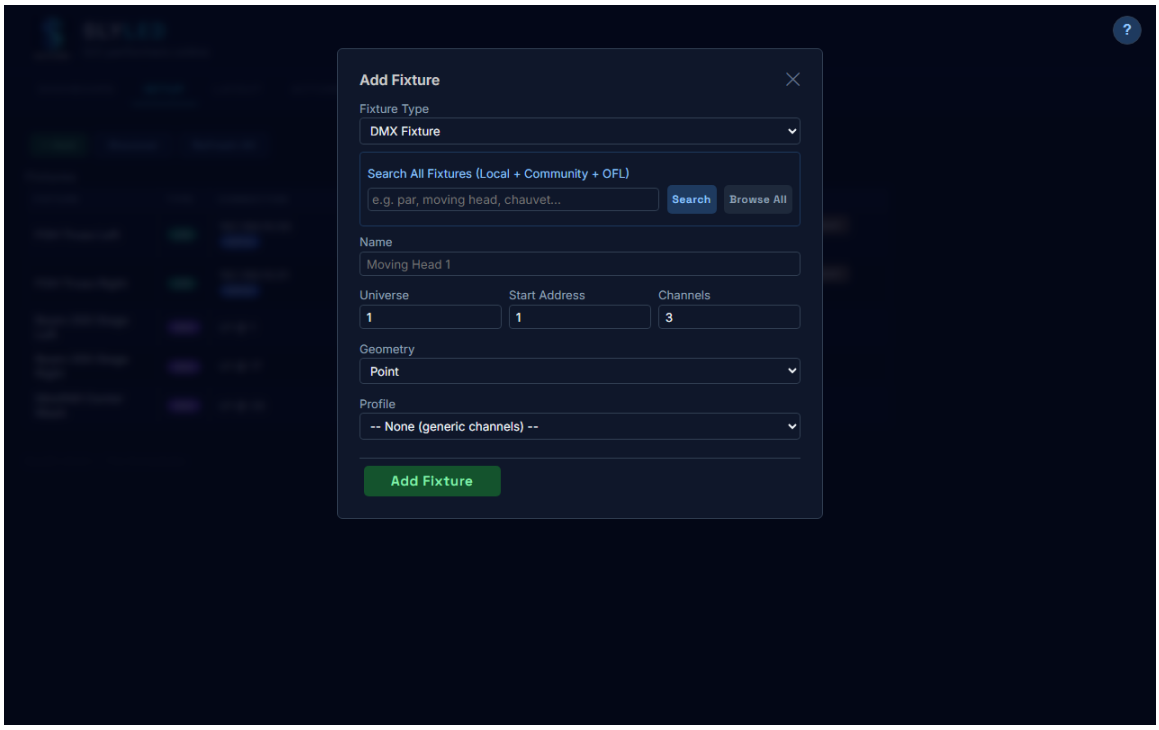
Add Fixture — LED performer flow with IP address entry

Adding a DMX Fixture

Click Add Fixture, select "DMX Fixture", and configure:

- Name: Descriptive label
- Universe: DMX universe number (1+)
- Start Address: DMX start channel (1-512)
- Channel Count: Number of channels the fixture uses
- Profile: Select from the built-in library or import from Open Fixture Library

DMX fixtures appear in the setup table with purple "DMX" badges.

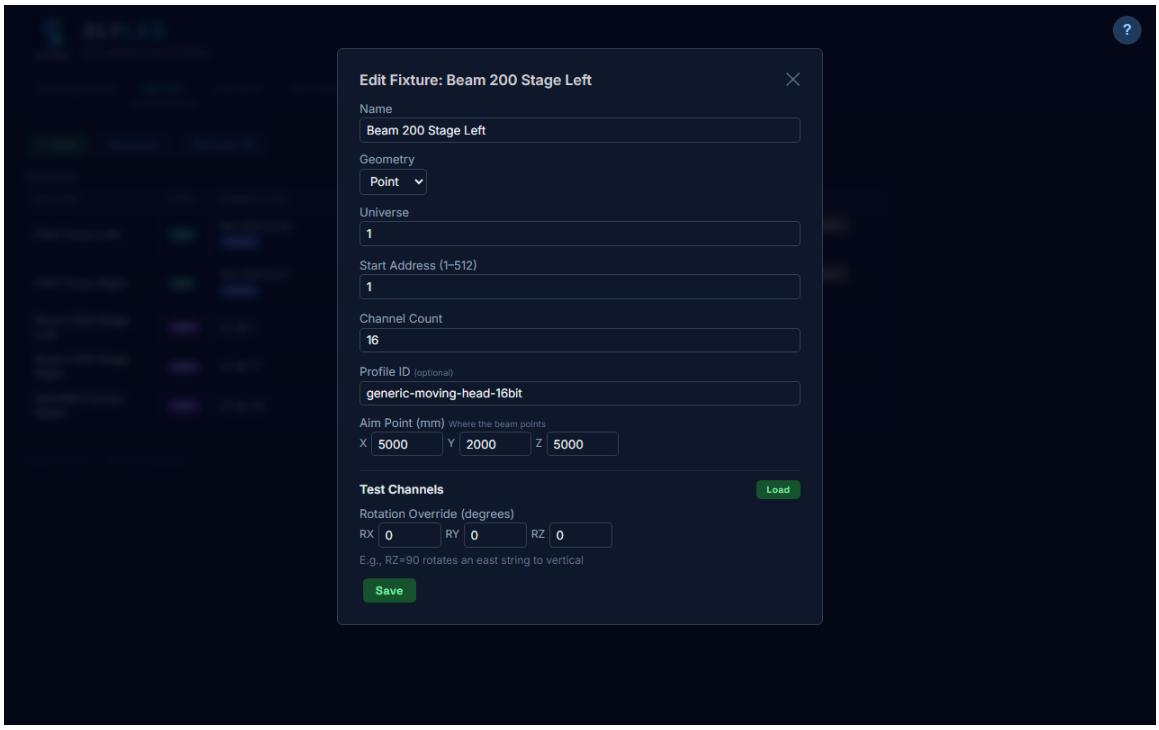


Add Fixture — DMX fixture with profile dropdown

Editing DMX Fixtures

Click Edit on a DMX fixture to modify its properties. The edit modal shows:

- Universe, Start Address, Channel Count
- Profile ID (for channel name/type mapping)
- Aim Point (X, Y, Z in mm) — where the beam points for moving heads
- Test Channels — interactive sliders to control each DMX channel live
- Rotation Override (degrees)



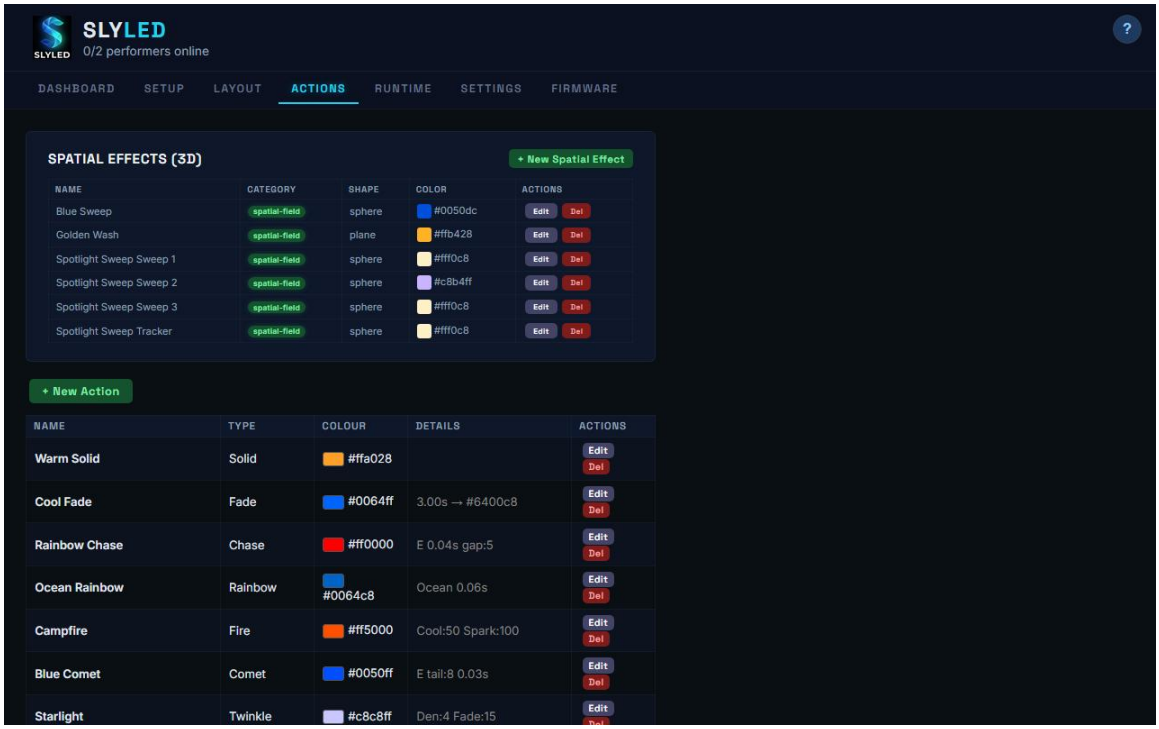
Edit DMX fixture — aim point, profile, and test channels

Fixture Types

Type	Description	Use Case
Linear	LED strip/string with pixels along a path	LED performers
Point	Single light source with area of effect	DMX pars, spots, moving heads
Surface	3D mesh as a projection target	Walls, screens
Group	Named collection of fixtures	Zones, grouped control

3. Creating Spatial Effects

Spatial effects operate in 3D space. A sphere of colored light sweeping across the stage illuminates different fixtures at different times based on their physical position. For DMX moving heads, the effect center becomes the aim target — heads track the effect as it moves.



Actions tab — spatial effects and classic action library

Spatial Fields

Field	Description
Shape	Sphere, Plane, or Box
Color	RGB color applied to pixels inside the field
Size	Radius (sphere), thickness (plane), or W/H/D (box)
Motion Start	Starting position [x, y, z] in mm
Motion End	Ending position [x, y, z] in mm
Duration	Travel time from start to end
Easing	Linear, ease-in, ease-out, ease-in-out
Blend	Replace, Add, Multiply, or Screen

Moving Heads + Spatial Effects

When a spatial effect is applied to a DMX moving head fixture:

- The effect's center position becomes the aim target
- Pan/tilt angles are computed from fixture position to aim point

- Color is applied from the effect's RGB values
- Moving effects produce time-sliced segments that track the motion
- 3D viewport shows beam cones pointing at the effect center

DMX Action Types

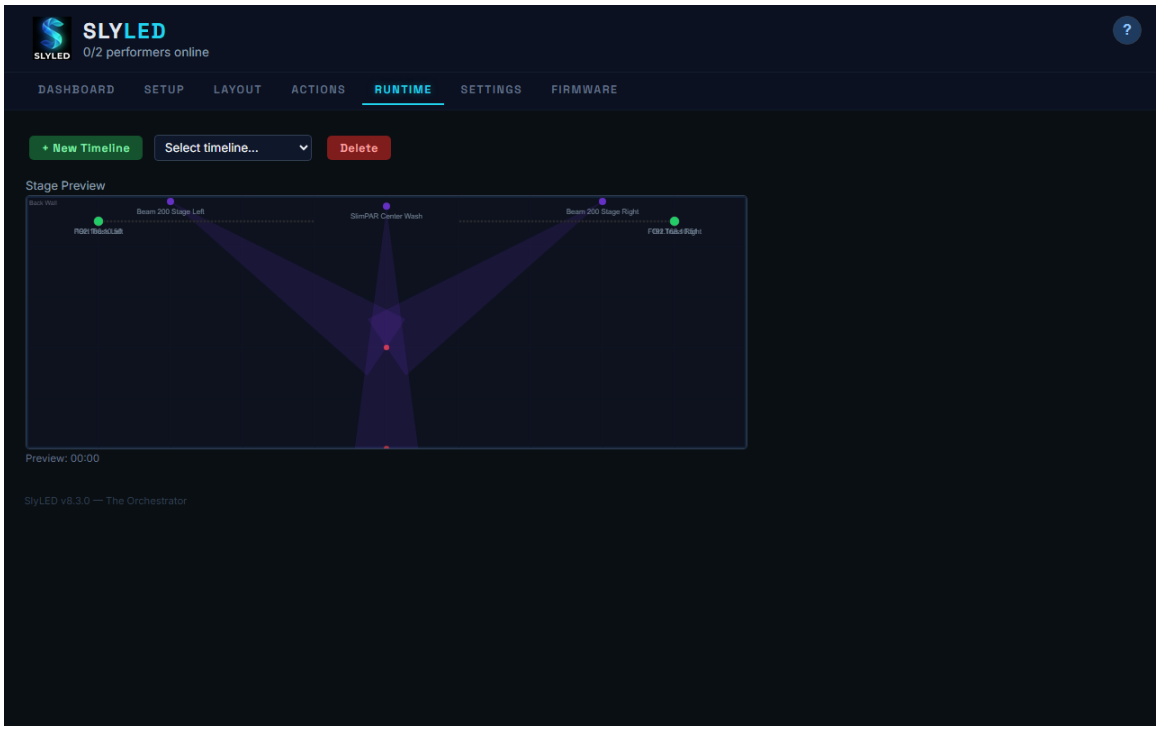
In addition to classic LED actions (Solid, Chase, Rainbow, etc.), four DMX-specific action types are available for direct control of DMX features:

Action Type	Description
DMX Scene	Set exact values for dimmer, pan, tilt, strobe, gobo, color wheel, prism
Pan/Tilt Move	Animate pan/tilt from start to end position over time
Gobo Select	Select a gobo wheel position with optional color
Color Wheel	Select a color wheel position

Classic LED actions (Solid, Breathe, Chase, etc.) are automatically converted to DMX Scene segments when assigned to DMX fixtures. The dimmer is auto-set to 255 when color is active, and pan/tilt default to center (0.5).

4. Building a Timeline

Timelines are multi-track, overlapping effect sequences with precise timing. Use the Runtime tab to create and edit timelines.



Runtime tab — timeline editor

Creating a Timeline

1. Click "+ New Timeline" and enter name and duration
2. Select the timeline from the dropdown
3. Add tracks (one per fixture or "All Performers")
4. Add clips referencing spatial effects or classic actions
5. Adjust clip timing by editing start time and duration

5. Baking & Playback

What Is Baking?

Baking compiles a timeline into minimal action instructions for each performer. The smart bake engine analyzes each clip's spatial geometry directly and computes per-string sweep patterns, directions, and speeds.

For DMX fixtures, baking also computes:
- Pan/tilt angles from spatial effect motion paths
- Color values from the effect's intersection with the fixture position
- Dimmer values (auto 255 when color is active)
- Time-sliced segments for smooth moving head tracking

Playback

After baking and syncing:
- LED performers receive action steps via UDP and execute locally
- DMX fixtures are driven by a 40Hz playback loop sending Art-Net/sACN packets
- All channels (RGB, pan, tilt, dimmer, strobe, gobo) are sent per profile
- 16-bit channels (pan/tilt) are split into coarse + fine bytes automatically

6. Show Preview Emulator

The emulator shows a real-time preview of your show on the Runtime tab.

- LED fixtures: Colored dots along string directions, animated per-pixel
- DMX fixtures: Beam triangles from fixture position toward aim point
- Beam color and alpha reflect the fixture's current DMX state
- Time counter shows elapsed position

7. Preset Shows

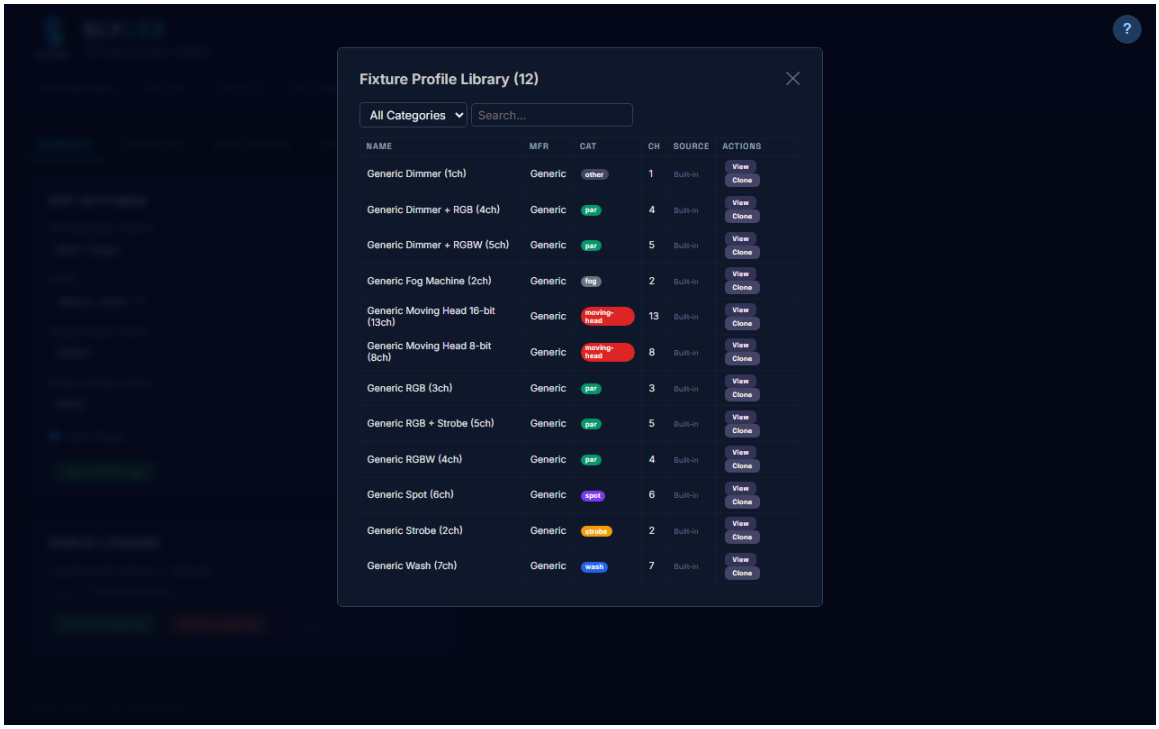
14 themed shows are available from the Runtime tab. Shows are dynamically generated based on your actual fixtures — every fixture gets coverage with no dark periods. LED fixtures get pattern effects, DMX pars get color washes, and moving heads get pan/tilt sweeps that track spatial effects across the stage.

Each load produces a unique variation of the theme with randomized timing and positions.

Preset	Type	Description
Rainbow Up	Spatial plane	Rainbow from floor to ceiling
Rainbow Across	Spatial sphere	Rainbow sweeping left to right
Slow Fire	Classic action	Warm fire effect on all fixtures
Disco	Classic action	Pastel twinkle sparkles
Ocean Wave	Spatial (2 effects)	Blue wave sweep with teal wash
Sunset Glow	Mixed	Warm breathe with golden plane sweep
Police Lights	Mixed	Red strobe with blue box flash sweep
Starfield	Classic action	White sparkles on dark background
Aurora Borealis	Spatial (2 effects)	Green curtain with purple shimmer
Spotlight Sweep	Spatial (moving heads)	Warm orb sweeps stage — heads track it
Concert Wash	Mixed (moving heads)	Magenta flood + amber tracking spot
Figure Eight	Spatial (moving heads)	Crossing orbs — heads trace X paths
Thunderstorm	Mixed (moving heads)	Lightning strikes — heads chase bolts
Dance Floor	Mixed (moving heads)	Fast orbiting spots — rapid tracking

8. DMX Fixture Profiles

Profiles define the channel layout and capabilities of DMX fixtures. Each channel maps a DMX offset to a function (pan, tilt, red, green, blue, etc.) with capability ranges that describe what each DMX value range does.



Fixture Profile Library — 12 built-in profiles

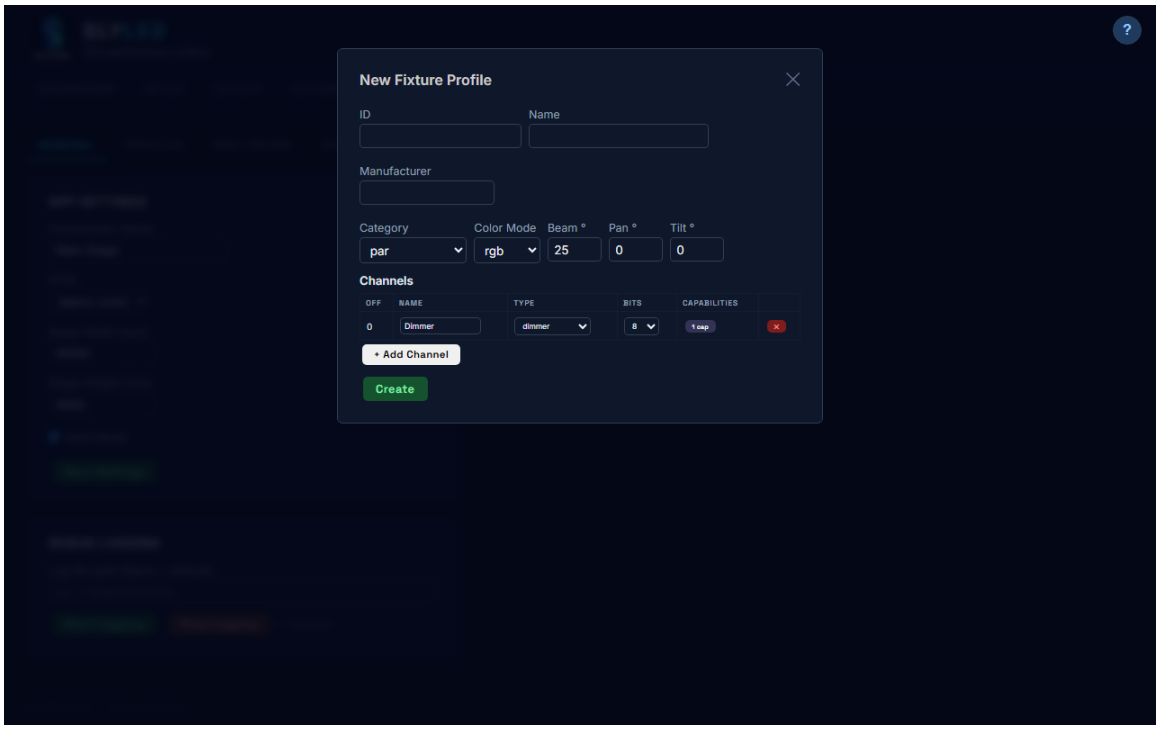
Viewing a Profile

Click View on any profile to see the full channel table with capabilities. Moving head profiles show pan/tilt ranges, beam width, and color mode.

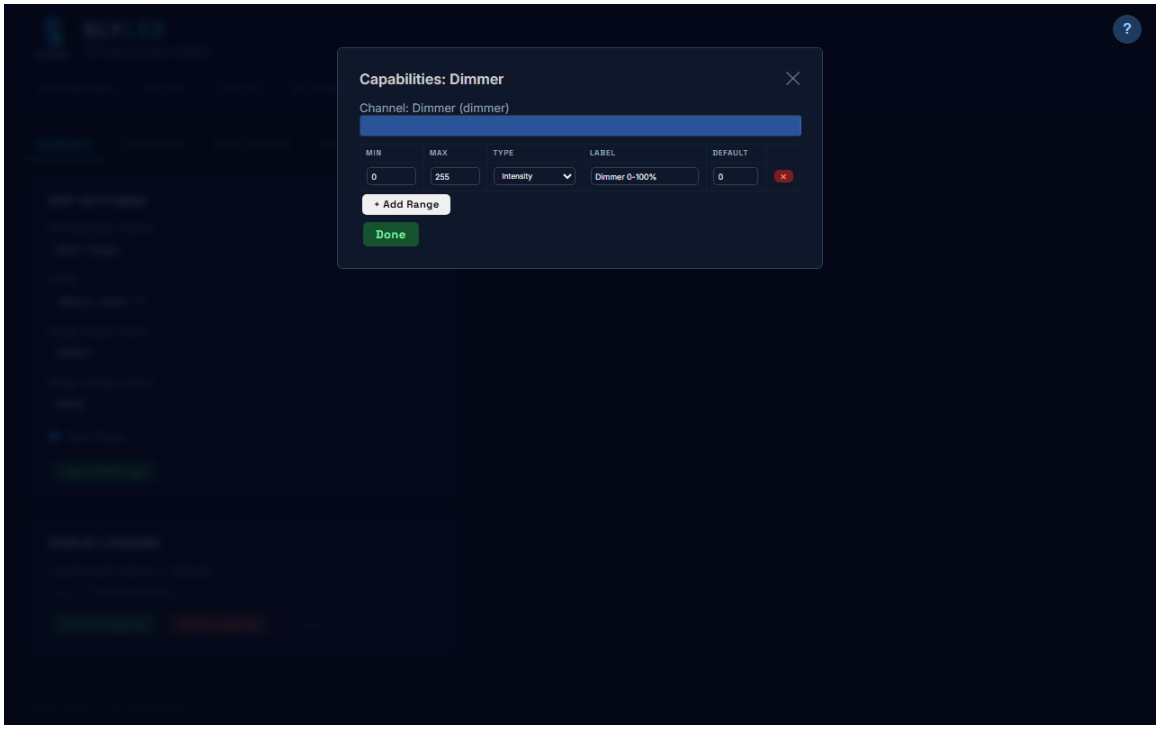
Moving Head 16-bit profile with 13 channels and capability ranges

Creating Custom Profiles

Click "New Profile" to open the editor. Define channels with type, name, bits (8/16), and capability ranges. Click the capabilities button on each channel to define what DMX value ranges do (e.g., gobo positions, strobe speeds, color wheel slots).



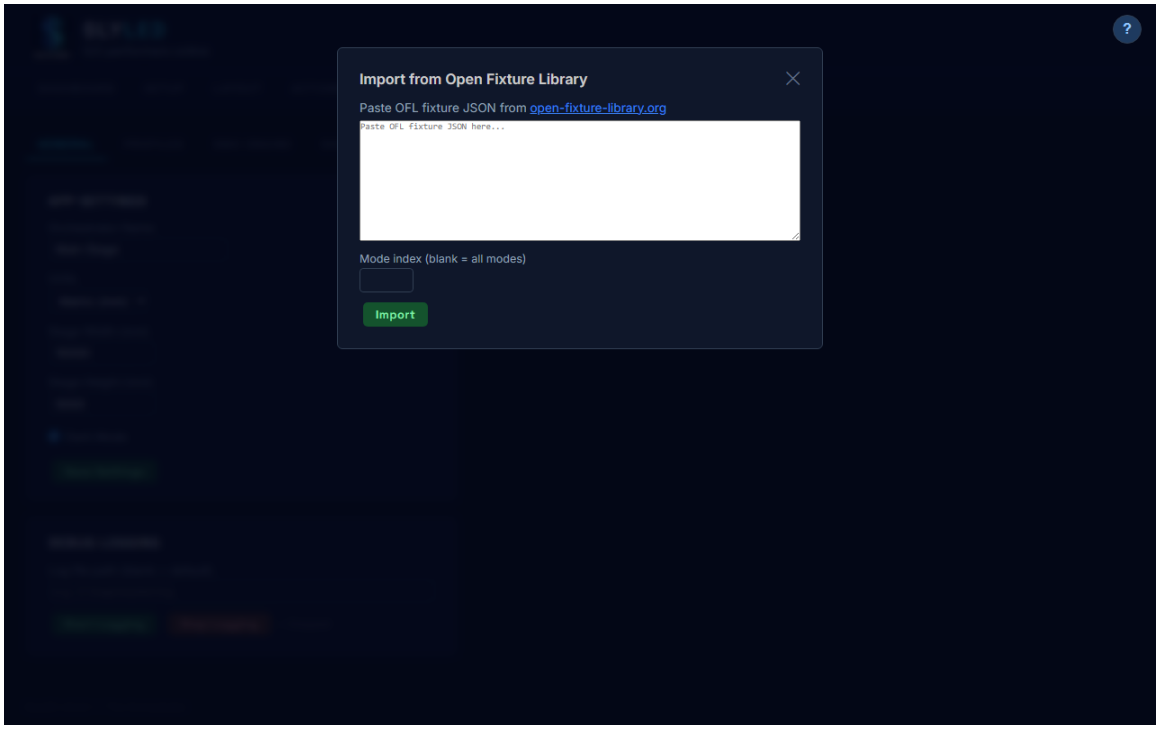
Profile editor — channel table with type dropdowns



Capability range editor for a channel

Importing from Open Fixture Library

Click "Import OFL" to paste fixture JSON from open-fixture-library.org. The importer converts OFL's capability model to SlyLED format, handling multi-mode fixtures, 16-bit channels, and color detection automatically.

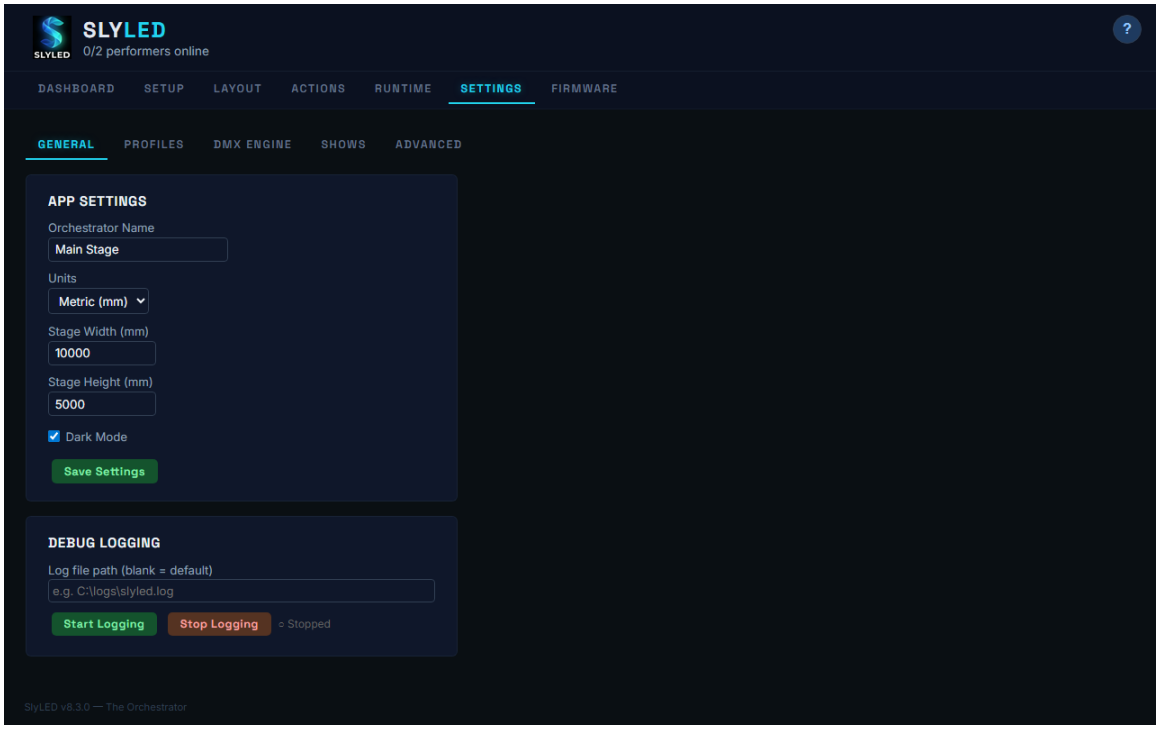


Open Fixture Library import modal

Import/Export Bundles

Export your custom profiles as a JSON bundle file for backup or sharing. Import bundles to add profiles from other installations.

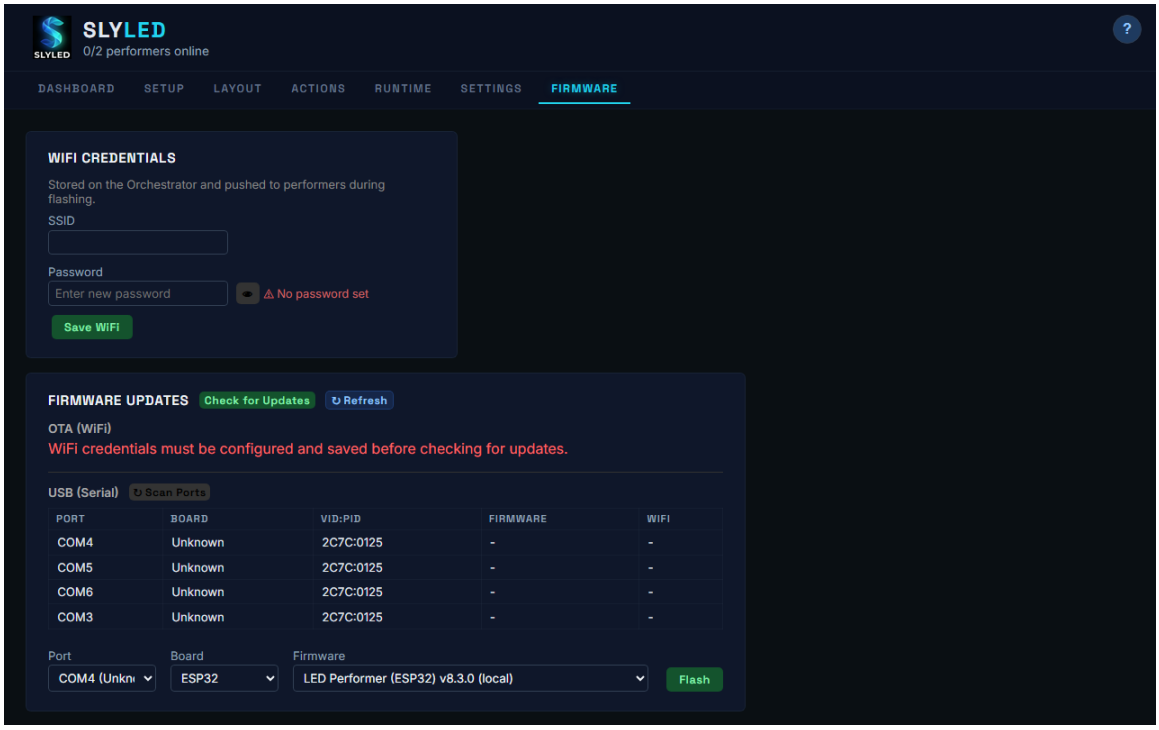
9. Settings & Configuration



Settings tab — app configuration and DMX output

The Settings tab contains:
- Orchestrator name and units
- Stage dimensions
- Dark mode toggle
- Logging control
- Configuration export/import
- Show export/import
- DMX output settings (Art-Net/sACN, frame rate, universe routing)
- Fixture Profile Library
- Factory reset

Firmware



Firmware tab — board detection and flash controls

The Firmware tab provides:

- COM port detection with board type identification
- Serial version query
- WiFi credential management
- One-click firmware flashing for ESP32, D1 Mini, and Giga R1 boards

10. System Limits

Resource	Limit	Notes
Children (performers)	8 max	Protocol constant
Strings per child	8 max	ESP32 supports up to 8 GPIO pins
LEDs per string	65535 max	uint16_t addressing (protocol v4)
Steps per runner	16 max	LoadStepPayload array limit
Bake segments per fixture	64 max	Supports PT move time slices
Bake frame rate	40 Hz	Art-Net output rate
DMX universes	Unlimited	One Art-Net packet per universe
DMX channels per universe	512	DMX-512 standard
Pan/tilt resolution	8 or 16 bit	Per profile channel definition
Preview resolution	1 fps	1 color per string per second
NTP sync offset	5 seconds	GO command sent with future epoch

11. Troubleshooting

3D Viewport Not Rendering

Browser doesn't support WebGL. Use Chrome, Firefox, or Edge.

Performers Not Syncing

Children offline or on different network. Check Setup tab status.

Bake Error: "No fixtures"

Add and position fixtures in Layout before baking.

DMX Not Outputting

Check Art-Net engine is started (Settings → DMX Output → Start). Verify universe routing matches your bridge address.

Moving Heads Not Tracking

Verify fixture has a profile with panRange/tiltRange > 0. Check aimPoint is set (Edit fixture → Aim Point fields).

Preview Shows No DMX Beams

Bake the timeline first. DMX preview requires baked data.

Factory Reset

Settings → Factory Reset clears ALL data including fixtures, effects, timelines, and profiles.

12. API Quick Reference

All endpoints are served from the Orchestrator at http://localhost:8080.

Fixtures

Method	Endpoint	Description
GET/POST	/api/fixtures	List / create
GET/PUT/DELETE	/api/fixtures/:id	CRUD
PUT	/api/fixtures/:id/aim	Set DMX aim point
POST	/api/fixtures/:id/resolve	Compute pixel positions

DMX Profiles

Method	Endpoint	Description
GET/POST	/api/dmx-profiles	List / create
GET/PUT/DELETE	/api/dmx-profiles/:id	CRUD
GET	/api/dmx-profiles/export	Export bundle
POST	/api/dmx-profiles/import	Import bundle
POST	/api/dmx-profiles/ofl/import-json	Import OFL fixture

Timelines

Method	Endpoint	Description
GET/POST	/api/timelines	List / create
POST	/api/timelines/:id/bake	Start baking
POST	/api/timelines/:id/start	Start playback
POST	/api/timelines/:id/stop	Stop playback

DMX Output

Method	Endpoint	Description
GET	/api/dmx/status	Engine status
POST	/api/dmx/start	Start engine
POST	/api/dmx/stop	Stop engine
GET	/api/dmx/fixture/:id/channels	Fixture channels with values